

$$2) X = \begin{bmatrix} 0.34 \\ 0.55 \end{bmatrix}, W^1 = \begin{bmatrix} 0.42 & 0.72 \\ 0 & 0.3 \\ 0.15 & 0.09 \end{bmatrix}, W^2 = \begin{bmatrix} 0.44 & 0.03 \end{bmatrix}$$

$$W^2 = \begin{bmatrix} 0.44 & 0.03 & 0.55 \end{bmatrix}, W^3 = \begin{bmatrix} 0.55 \\ 0.71 \\ 0.29 \end{bmatrix}$$

$$W^4 = \begin{bmatrix} 0.97 & 0.55 & 0.97 \\ 0.71 & 0.7 & 0.22 \end{bmatrix} \quad LR = 0.5$$

$$L_1 = \begin{bmatrix} 0.42 & 0.72 \\ 0 & 0.3 \\ 0.15 & 0.09 \end{bmatrix} \begin{bmatrix} 0.34 \\ 0.55 \end{bmatrix} = \begin{bmatrix} 0.5388 \\ 0.1650 \\ 0.1005 \end{bmatrix}$$

$$L_2 = \begin{bmatrix} 0.44 & 0.03 & 0.55 \end{bmatrix} \begin{bmatrix} 0.5388 \\ 0.1650 \\ 0.1005 \end{bmatrix} = \begin{bmatrix} 0.2973 \end{bmatrix}$$

$$L_3 = \begin{bmatrix} 0.55 \\ 0.71 \\ 0.29 \end{bmatrix} \begin{bmatrix} 0.2973 \end{bmatrix} = \begin{bmatrix} 0.1641 \\ 0.2111 \\ 0.0862 \end{bmatrix}$$

$$L_4 = \begin{bmatrix} 0.97 & 0.55 & 0.97 \\ 0.71 & 0.7 & 0.22 \end{bmatrix} \begin{bmatrix} 0.1641 \\ 0.2111 \\ 0.0862 \end{bmatrix} = \begin{bmatrix} 0.3007 \\ 0.2406 \end{bmatrix}$$

$$\text{Reconstruction Loss} = -(0.34 \ln(0.3007) + 0.55 \ln(0.2406)) \\ = 1.1921$$

Start BackProp:-

$$dZ^4 = \begin{bmatrix} 0.3007 \\ 0.2406 \end{bmatrix} - \begin{bmatrix} 0.34 \\ 0.55 \end{bmatrix} = \begin{bmatrix} -0.0393 \\ -0.3094 \end{bmatrix}$$

$$dw^4 = \begin{bmatrix} -0.0393 \\ -0.3094 \end{bmatrix} \cdot \begin{bmatrix} 0.1041 \\ 0.2111 \\ 0.0862 \end{bmatrix}$$

Switch around matrix position and transpose dZ^4

~~$$dw^4 = \begin{bmatrix} 0.1041 \\ 0.2111 \\ 0.0862 \end{bmatrix} \begin{bmatrix} -0.0393 & -0.3094 \end{bmatrix}$$~~
~~$$= \begin{bmatrix} -0.00481 & -0.0322 \\ -0.0083 & -0.0653 \\ -0.0034 & -0.0267 \end{bmatrix}$$~~

$$dZ^3 = \begin{bmatrix} 0.97 & 0.71 \\ 0.55 & 0.7 \\ 0.97 & 0.22 \end{bmatrix} \cdot \begin{bmatrix} -0.0393 \\ -0.3094 \end{bmatrix} = \begin{bmatrix} -0.2578 \\ -0.2382 \\ -0.1062 \end{bmatrix}$$

$$dw^3 = \begin{bmatrix} -0.2578 \\ -0.2382 \\ -0.1062 \end{bmatrix} \begin{bmatrix} 0.2973 \end{bmatrix} = \begin{bmatrix} -0.0766 \\ -0.0708 \\ -0.0316 \end{bmatrix}$$

$$d_2^2 = \begin{bmatrix} 0.35 & 0.71 & 0.29 \end{bmatrix} \begin{bmatrix} -0.2578 \\ -0.2382 \\ -0.1062 \end{bmatrix} = \begin{bmatrix} -0.2902 \end{bmatrix}$$

$$dw^2 = \begin{bmatrix} -0.2902 \end{bmatrix} \begin{bmatrix} 0.5388 & 0.1650 & 0.1005 \end{bmatrix} = \begin{bmatrix} -0.1564 & -0.0479 & -0.0292 \end{bmatrix}$$

$$d_2^1 = \begin{bmatrix} 0.44 \\ 0.03 \\ 0.55 \end{bmatrix} \begin{bmatrix} -0.2902 \end{bmatrix} = \begin{bmatrix} -0.1277 \\ -0.0087 \\ -0.1596 \end{bmatrix}$$

$$dw^1 = \begin{bmatrix} -0.1277 \\ -0.0087 \\ -0.1596 \end{bmatrix} \begin{bmatrix} 0.34 & 0.55 \end{bmatrix} = \begin{bmatrix} -0.0434 & -0.0702 \\ -0.0030 & -0.0048 \\ -0.0543 & -0.0878 \end{bmatrix}$$

Now update weights:-

$$W_{new}^4 = W_{old}^4 - 0.5(dw^4)$$

$$= \begin{bmatrix} 0.97 & 0.55 & 0.97 \\ 0.71 & 0.7 & 0.22 \end{bmatrix} - 0.5 \begin{bmatrix} -0.0041 & -0.0322 \\ -0.0083 & -0.0653 \\ -0.0034 & -0.0257 \end{bmatrix}$$

Transpose dw_4

$$dw_4 = \begin{bmatrix} -0.0393 \\ -0.3099 \end{bmatrix} \begin{bmatrix} 0.1041 & 0.2111 & 0.862 \end{bmatrix}$$

$$= \begin{bmatrix} -0.0041 & -0.0083 & -0.0034 \\ -0.0032 & -0.0653 & -0.0267 \end{bmatrix}$$

∴ Update weights:-

$$W_{new}^{[4]} = W_{old}^{[4]} - d(dw^{[4]})$$

$$W_{new}^{[4]} = \begin{bmatrix} 0.97 & 0.55 & 0.97 \\ 0.71 & 0.7 & 0.22 \end{bmatrix} - 0.5 \begin{bmatrix} -0.0041 & -0.0083 & -0.0034 \\ -0.0032 & -0.0653 & -0.0267 \end{bmatrix}$$

$$= \begin{bmatrix} 0.97205 & 0.55415 \end{bmatrix}$$

$$= \begin{bmatrix} 0.97205 & 0.55415 & 0.9727 \end{bmatrix}$$

$$= \begin{bmatrix} 0.9722 & 0.5542 & 0.9717 \\ 0.7261 & 0.7327 & 0.2334 \end{bmatrix}$$

$$W_{new}^{[3]} = \begin{bmatrix} 0.35 \\ 0.71 \\ 0.29 \end{bmatrix} - (0.5) \begin{bmatrix} -0.0766 \\ -0.0708 \\ -0.0316 \end{bmatrix}$$

$$= \begin{bmatrix} 0.3883 \\ 0.7454 \\ 0.3058 \end{bmatrix}$$

$$W_{new}^2 = [0.44 \ 0.03 \ 0.55] - 0.5 [-0.1564 \ -0.0479 \ -0.2922]$$

$$= [0.5192 \ 0.0540 \ 0.5646]$$

$$W_{new}^1 = \begin{bmatrix} 0.42 & 0.72 \\ 0 & 0.3 \\ 0.15 & 0.09 \end{bmatrix} - (0.5) \begin{bmatrix} -0.0434 & -0.0702 \\ -0.0030 & -0.0048 \\ -0.0543 & -0.0878 \end{bmatrix}$$

$$= \begin{bmatrix} 0.4417 & 0.7551 \\ 0.0015 & 0.3024 \\ 0.1772 & 0.1339 \end{bmatrix}$$

3- ~~3~~ All initial weights are 0.5

$$\therefore W_1, W_2, W_3, W_4 = 0.5$$

$$L_1 = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.34 \\ 0.55 \end{bmatrix} = \begin{bmatrix} 0.445 \\ 0.445 \\ 0.445 \end{bmatrix}$$

$$L_2 = \text{latent space} = 2 = \mu + \sigma \odot \epsilon$$

$$\mu = \begin{bmatrix} 0.5 & 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.445 \\ 0.445 \\ 0.445 \end{bmatrix} = [0.6675]$$

$$\log(0^2) = [0.5 \ 0.5 \ 0.5] \begin{bmatrix} 0.445 \\ 0.445 \\ 0.445 \end{bmatrix} = [0.6675]$$

$$\sigma = e^{0.6675 \times 0.5} = 1.40, \text{ Assume } \epsilon = 0.5$$

$$\therefore z = 0.6675 + 1.40 \times 0.5 = 1.37$$

$$L_3 = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \end{bmatrix} [0.6675 + 1.37] = \begin{bmatrix} 0.685 \\ 0.685 \\ 0.685 \end{bmatrix}$$

$$L_4 = \begin{bmatrix} 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.685 \\ 0.685 \\ 0.685 \end{bmatrix} = \begin{bmatrix} 1.0275 \\ 1.0275 \end{bmatrix}$$

What's ~~needed~~ ^{needed} for back prop:-

Reconstruction Loss and KL divergence term

$$RL = \sum (u_i - \hat{u}_i)^2$$

KLDs

$$Loss = -E_{z \sim q_\phi} [\ln p_\theta(x|z)] + D_{KL}(q_\phi(z|x) \| p_\theta(z))$$

Derivatives:-

Since the z mean and z -std ~~are~~ are used for reading z , their derivatives are needed for backward pass.

$$\therefore \frac{dz}{d\mu} = 1, \quad \frac{dz}{d\sigma} = \epsilon$$

Epsilon is a constant so no need for derivation.

Derivative of KLD also needed:-

$$KLD = -\frac{1}{2} E(1 + \log(\sigma^2) - \mu^2 - \sigma^2)$$

$$\frac{dKLD}{d\mu} = \mu, \quad \frac{dKLD}{d\sigma} = \sigma - \frac{1}{\sigma}$$

~~These~~ These are all derivations unique to VAEs. The rest are already used by regular AEs like reconstruction derivative so not mentioned.